

	the distance 3.0 m is constant, the upward force by pivot on beam decreases.
4(a)(i)	Frictional force of the road on the bike Forward direction
4(a)(ii)	Viscous force / Air resistance / Drag Backward direction
4(b)	Distance travelled by bicycle $= \frac{1}{2}(8 + 10)(2) + (10)(20) + \frac{1}{2}(10)(5) = 243 \text{ m}$
5(a)	Total resistance of circuit